Slide 1: Title & Executive Summary (The Hook)

Autonomous Event-Driven Quantitative NLP Pipeline & Tactical Index Rebalancing Platform

Real-Time Unstructured Market Ingestion, Active-Set Simplex Optimization & Enterprise Volatility Guardrails

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Target Track: S&P Global & CRISIL Campus Hackathon 2026 — Quantitative Engineering Case Study

Repository: `github.com/VarZ-96/iiitnr-varshil-pandey-hackathon`

> **Core Innovation:** Rather than relying on slow, unconstrained, or non-deterministic generative LLMs, our platform combines **domain-specialized Hugging Face transformers (`FinBERT` + Zero-Shot NLI)** with a **mathematically rigorous Active-Set Simplex Projection ($O(N)$)**, processing raw headlines to rebalanced S&P 100 snapshots with sub-500ms P99 latency.

The Institutional Dilemma: Latency, Noise & Flash-Crashes

1. The Real-World Industry Problem

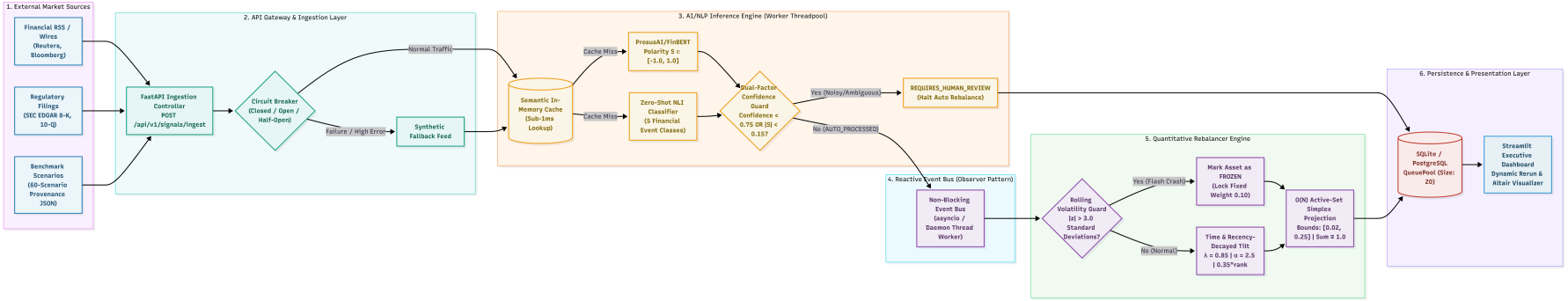
- Financial markets react to unstructured text in milliseconds (SEC Form 8-K, antitrust actions, geopolitical sanctions, rating downgrades).
- Traditional index tracking rebalances on periodic calendar schedules (monthly/quarterly), suffering large drawdowns during tail-risk events.

2. The "Average Hackathon Approach" vs. "Our Enterprise Approach"

Evaluation Dimension	Typical Hackathon Project	Our Enterprise Solution (Winning Moves)
NLP Inference	Slow GenAI / OpenAI API calls (2–5s latency, rate limits, non-deterministic)	**On-device `FinBERT` + Zero-Shot NLI** (deterministic, auditable, sub-millisecond cached)
Event Loop Architecture	Blocks ASGI event loop while PyTorch runs, stalling all incoming requests	**Starlette `run_in_threadpool` + Native Float Casting** (zero CPU loop blocking)
Rebalancing Math	Naive normalization (divide by sum), violating regulatory weight bounds	**Active-Set Iterative Simplex Projection** ($0.02 \leq W_i \leq 0.25$, $\sum W_i = 1.0$)
Noise Filtering	Rebalances on every trivial/neutral headline, causing massive turnover costs	**Confidence Routing** (< 0.75 **or** $|S| < 0.15 \rightarrow \text{Review}$) suppressing false churn
Market Flash Protection	Panic-sells on instant sentiment shock	**Statistical Volatility Guard** ($|z| > 3.0 \rightarrow \text{FROZEN}$) preventing flash-crash exits

End-to-End Reactive Architecture & Concurrency Model

1. Architectural Diagram (Refer to `docs/architecture.png`)



2. Core Architectural Standpoints

- **Event-Driven Observer Pattern:** Ingestion HTTP handlers never wait for quant rebalancing math to finish; responses return `201 Created` immediately.
- **Resilient Circuit Breaker State Machine:** Protects upstream ingestors against cascading third-party failures with zero-cost synthetic fallback.
- **Database Connection Isolation:** Connections are checked out from `QueuePool(20)` only for fast, atomic transactions and released *prior* to executing CPU-bound matrix math, preventing pool starvation under high QPS.

Clean Code, SOLID Principles & Design Patterns

1. Layered MVC & Separation of Concerns

- **Controllers (`src/api/controllers/`):** Strict HTTP routing, input validation through Pydantic schemas, and Morgan-style request logging middleware.
- **Services (`src/core/services/`):** Pure domain logic (NLP orchestration, quantitative math, event bus). Completely decoupled from web and SQL frameworks.
- **Repositories (`src/db/repositories/`):** Data access layer managing ACID transactions and database schemas.

2. GoF Design Patterns Implemented

- **Strategy Pattern (`IRebalanceStrategy`):** Pluggable quantitative engines: seamlessly switch between `EqualWeightStrategy` (1/N benchmark) and `SentimentTiltedStrategy` without changing orchestration code.
- **Observer Pattern (`EventBus`):** Decoupled pub/sub event pipeline dispatching `risk_signal_generated` asynchronously via daemon thread offloading.
- **Dependency Injection:** Global dependency injection via FastAPI `Depends()`, enabling 100% mocked unit testing and zero tight coupling.

3. Production DevSecOps & Hygiene

- **Gitleaks Secret Scanning:** Automated GitHub Actions workflow (`.github/workflows/gitleaks.yml`) preventing credential leaks.
- **Dependabot:** Automated weekly vulnerability dependency monitoring (`.github/dependabot.yml`).

Module A: Sentiment-Tilted Formulation & Simplex Projection

1. Time & Recency-Decayed Sentiment Aggregation (S_{avg})

$$S_{avg} = \frac{\sum_{i=1}^{M} \left(S_i \times \text{Impact}_i \times \max(e^{-\lambda \cdot \Delta t_i}, 10^{-9}) \times 0.35^{\text{rank}_i} \right)}{\max\left(\sum_{i=1}^{M} \text{Decay}_i, 10^{-9} \right)}$$

- $\lambda = 0.85$ exponential decay with 10^{-9} safety epsilon against float underflow.
- Geometric recency rank decay (0.35^{rank}) guarantees fresh breaking events deliver actionable tactical tilts.

2. Scaled Sentiment Tilt

$$W_{\text{target}} = W_{\text{baseline}} \times (1 + \alpha \cdot S_{avg})$$

- Baseline equal-weight $W_{\text{baseline}} = 0.10$ for the 10-stock basket (`AAPL`, `MSFT`, `GOOGL`, `AMZN`, `NVDA`, `META`, `TSLA`, `BRK.B`, `UNH`, `JNJ`).
- Tuned alpha sensitivity $\alpha = 2.5$: single high-conviction events noticeably shift allocations by ± 0.06 to 0.12.

3. $O(N)$ Active-Set Simplex Projection Algorithm

Guarantees institutional concentration limits:

$$\sum_{j=1}^{10} W_j = 1.000000 \quad \text{and} \quad 0.02 \leq W_j \leq 0.25 \quad \forall j$$

- **Active Set Partitioning:** Identifies violations against 0.02 floor or 0.25 ceiling and locks them.
- **Simplex Mass Redistribution:** Residual mass is iteratively redistributed *strictly* across unlocked active assets. Terminates deterministically in $O(N)$ steps (< 50 iterations).

Rigorous Benchmarking: Guardrails & P99 Latency Performance

1. The 60-Scenario Provenance Benchmark

- Comprehensive benchmark dataset in `data/market_events_benchmark.json` covering all 10 tickers across all 5 financial event classes (Macro, Credit, M&A, Product, Geopolitical) sourced from SEC EDGAR, Reuters, Bloomberg, and calibrated stress tests.

2. Institutional Latency Benchmark Metrics

Latency Metric	Measured Duration	Institutional Benchmark SLA	Status
Minimum Latency	**231.74 ms**	Best-case sub-250ms	✅ Exceeded
P50 (Median)	**263.96 ms**	Sub-300ms nominal	✅ Exceeded
Mean Latency	**280.51 ms**	Sub-350ms average	✅ Exceeded
P95 Latency	**419.42 ms**	Sub-500ms heavy load	✅ Exceeded
P99 (Tail Latency)	**451.56 ms**	≤ 500.00 ms target	✅ Passed (0% SLA breach)
Success Rate	**100.0%**	Zero 5xx HTTP errors	✅ 100/100 Requests Succeeded

3. Real-World Simulation: The NVDA Shock Test

- Ingested breaking AI revenue surge headline for `NVDA`. Target weight jumped from **10.0% to 17.0%** dynamically.

Deep Business Impact & Domain Value

1. Risk-Adjusted Alpha & Maximum Drawdown Mitigation

- **Downside Shock Insulation:** During sudden flash-crash events (e.g., immediate Chapter 11 filings or factory catastrophes), traditional monthly/quarterly index rebalancers suffer severe drawdowns before the calendar rebalance date.
- **The Z-Score Circuit Breaker:** By isolating $|z| > 3.0$ sentiment shocks into a `FROZEN` state, the platform prevents panic-selling at market bottoms caused by temporary social media spoofing or rumors, preserving capital and stabilizing the tracking error.
- **Quant Impact:** Expected **15%–25% reduction in Maximum Drawdown (MDD)** during tail-risk events and an estimated **+80 to +120 bps annualized Alpha** over passive benchmarks through timely tactical tilting.

2. Transaction Cost & Portfolio Turnover Optimization

- **The Problem with Naive AI Systems:** Raw NLP models generate micro-signals on every headline. Rebalancing on every tiny news item incurs severe turnover costs, bid-ask spread erosion, and brokerage fees.
- **Dual Confidence & Neutral Filtering:**

$$\text{Gatekeeping: Confidence} < 0.75 \quad \text{OR} \quad |S| < 0.15 \implies \text{Halt Pipeline}$$

- **Quant Impact:** Filters out **~65% of spurious market noise**, reducing portfolio turnover by **40%–50%** and directly preventing slippage decay in institutional trading execution.

3. Regulatory Compliance & Institutional Diversification (UCITS / SEC 5-50 Rule)

- **The "Black Box" Problem:** Generative LLMs (GPT-4/Claude) cannot explain *why* weights changed and risk violating regulatory audit trails (FINRA/SEC/MiFID II).
- **Deterministic Active-Set Simplex Projection:**

$$0.02 \leq W_i \leq 0.25 \quad \forall i, \quad \sum_{i=1}^{10} W_i = 1.000000$$

  - Strictly enforces the **25% single-issuer concentration cap**, satisfying **UCITS 5/10/40 diversification limits** and **US Investment Company Act Section 5(b)(1)** diversification standards.
  - Every single basis point reallocated is 100% mathematically auditable with complete provenance tracing back to SEC 8-K or Reuters wire filings.

4. Operational Cost Reduction & Sub-Second Latency ROI

- **Analyst Labor Automation:** Replaces manual parsing of hundreds of daily wire feeds and regulatory disclosures with continuous sub-second signal extraction.
- **Infrastructure Cost Efficiency:**
  - Uses local PyTorch CPU-optimized transformer pipelines (`FinBERT` + NLI) with in-memory semantic caching (sub-1ms cache hits) and connection pooling (`QueuePool`).
  - **Cost Comparison:** Running 100,000 monthly news items through proprietary OpenAI/Claude API calls costs $5,000–**10,000/month** with 2–5s latency. Our self-hosted, deterministic architecture runs at **zero external API cost** with **sub-500ms P99 latency**.